

Final Project theory

① Explain mean, median, mode in context of customer income?

Ans:- mean is average income
median is middle income value, when income are arranged in order.
mode is most common income.

② Differentiate between standard deviation and Variance using loan amount?

Ans:- Variance:- measure how much loan amount spread out from the average. $\text{im-re}^{\text{square unit}}$
Standard deviation:- measure how much loan amount spread out from average in same unit, it is square root of variance.

③ What is random variable? give one example from the dataset?

Ans:- A value that changes randomly and unpredictably.

Ex income is random variable, as each changes for all customers.

④ Explain conditional probability in term of loan defaults?

Ans:- conditional probability of loan defaults is likelihood of customer defaulting, given specific condition, like having low credit score.

⑤ Define Bayes theorem and mention how banks can apply it?

Ans:- Bayes theorem calculate probability of an event based on prior knowledge and new evidence.

* Banks use it to predict loan defaults by updating risk assessment with customer data like credit score.

⑥ Difference between empirical and theoretical probability with examples?

Ans:- Theoretical probability give probability of assumption.

Ex Fair a coin, $P(\text{head}) = 0.5$

* Empirical probability give probability of actual experiments.

Ex Fair a coin, $P(\text{head}) = 0.7$

⑦ What is poisson distribution? give business example?

Ans:- poisson distribution measure probability of events happening in fixed interval.

Ex predict ^{no. of} customer arrival at store in an hour.

⑧ write a short note on eigenvalues and eigenvectors in data analysis?

Ans:- eigenvalue is scalar that represent how much matrix stretch or shrink a corresponding eigenvector.

* eigenvector that changes in only length, not in direction, when you multiply matrix with eigenvalue and its length is depend on eigenvalue.